An Analysis of Consumer Behavior in the Brazilian E-commerce Market: An Investigation of Customer Loyalty, Seasonality, and Regional Preferences

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Abstract

This paper investigates key drivers of consumer behavior within Brazil's emerging e-commerce market by analyzing the Olist Public Dataset. As technology increasingly allows for data-driven marketing strategies, understanding the nuanced behaviors of diverse consumer bases, especially in an emerging market like Brazil, is critical for firm success. This study tests three hypotheses regarding the relationship between positive customer reviews and e-loyalty, the fluctuation of seasonal sales during holiday periods, and the influence of geographic urbanization on product preferences. Utilizing regression analyses on 100,000 orders from 2016 to 2018, the results showed no significant relationship between positive review scores and repeat purchasing behavior, revealing an overwhelmingly transactional market where 99.4% of customers are one-time buyers. Second, the study found no significant difference in the proportion of hedonic purchases during holiday seasons compared to non-holiday months. Finally, contrary to expectations, urbanization was a significant negative predictor of hedonic consumption ($B = -0.002$, $p = .012$), explaining 22.5% of the variance in the hedonic ratio ($F(1, 25) = 7.265$, $p = .012$). This suggests rural consumers rely more on e-commerce to access luxury goods unavailable in their local physical markets. Further suggestions are made, including future studies' direction, and actionable insights for managers based on the findings.

Keywords

consumer behavior, marketing analysis, customer segmentation, Brazil

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Comments

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MGT 405: Advanced Statistical Methods for Business

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Abstract

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The evolution of technology has cemented marketing as a critical function for firm success, influencing financial performance, customer relationships, and new product performance (Moorman & Rust, 1999). The core value of marketing is determined by its ability to connect the consumer to the product and build essential market knowledge. In addition, the importance of marketing is further underscored by its significant correlation with company growth (Urbonavičius & Dikčius, 2008). With the rise of online shopping in the digital age, understanding the nuances of consumer behavior has become even more of a challenge, as businesses cannot operate without profits, which ties directly to customers, making customer satisfaction the ultimate goal of all business operations. Therefore, it is not over exaggerating to say that the market is transitioning into the phase "consumers are the king of the market" (Sundareswaran et al., 2022, p. 82). Hence, the birth of companies that focus heavily on the research of consumers/market, such as NielsenIQ, Ipsos, Kantar, etc. This paper will go deeper into analyzing consumer behavior of a Brazilian e-commerce platform, Olist, and whether or not certain patterns of the customers would result in them becoming frequent buyers, and how much the differences in socio-economics and culture affect consumers' behavior in shopping online in a developing country.

Nowadays, firms can easily collate and analyze vast amounts of data to uncover trends and patterns in consumer behavior, enabling the bridge between consumers and businesses, which helps organizations to create more individualized and precise marketing strategies (Liu et al., 2023). Specifically, the data-driven approach allows for enhanced customer relationship management by providing deep insights into customer behavior, preferences, and interactions.

By leveraging this information, companies can better identify consumer demands, recognize at-risk clients, and personalize interactions to improve engagement and retention (Odionu et al., 2024). Such targeted advertising is proven to be more persuasive and effective at influencing consumer attitudes and intentions than generic campaigns (Yeo et al., 2025). However, in order to maximize the increase in brand awareness, not only do companies have to curate the right advertisements to the right people, it had to be at the perfect timing as well (Bindah & Gunnoo, 2024).

One thing to note is that consumer behavior is not one size fits all. For example, in fashion, Khan et al. (2024) revealed that for Italian consumers, the main cultural dimension driving sustainable fashion purchasing is collectivism, while for Russian consumers, the cultural dimension driving sustainable fashion purchasing is long-term orientation. This application still persists when shopping is made accessible online: a study by Pratesi et al. (2021) highlights that a focus on personal autonomy (for individualistic cultures) and social recommendations (for collectivistic cultures) is essential for increasing online purchase intention, which confirms that culture is a significant determinant of consumer online purchase behavior. Therefore, in order to be successful, e-commerce strategies must be adapted to align with the dominant cultural dimensions of a target market.

The key characteristics in the emerging markets, like Brazil, include but are not limited to: the lack of digital infrastructure, mobile-first behavior, socio-cultural diversity, etc. (Adenuga, 2025). With such a niche market and the rare chance to access the data of e-commerce usage of the consumers in Brazil, this is a good opportunity for a data-driven analysis to understand more about the Brazilian demographics. This paper will examine three key pillars of the Brazilian digital consumer experience: customer loyalty, seasonal purchasing, and regional preferences through review scores, seasonality, and urbanization. By centering the analysis on these three factors, this study provides a comprehensive look at the interpersonal, temporal, and spatial drivers that shape the Brazilian digital consumer experience to provide empirical evidence for effective marketing in this landscape, all while using the Brazilian E-Commerce Public Dataset provided by Olist.

The selection of customer review scores, seasonality, and urbanization as the core independent variables for this study is based on the following: First, review scores are analyzed to evaluate the satisfaction-loyalty link of the consumers in a digital marketplace where buyers cannot physically inspect goods, high ratings serve as critical social proof that and validates seller reliability, which is theoretically expected to drive repeat purchasing behavior (Sudaryanto et al., 2025). Second, seasonality is included to investigate how cultural shopping windows such as Christmas or Black Friday influence the psychological transition from utilitarian necessity toward hedonic consumption, as these periods tend to have promotional surges that traditionally elevate the "wants" spending. Finally, urbanization is selected to capture the socio-economic and infrastructural divide across Brazil. Based on the premise that urban hubs often represent higher purchasing power and 'modern' lifestyle preferences, this variable allows for an examination of how geographic density and local market development dictate whether a consumer prioritizes pleasure-oriented (hedonic) or functional (utilitarian) goods (Sinha & Verma, 2019).

H1: Customers who leave positive reviews (4 or 5 stars) are more likely to become loyal customers to that store, defined as making one or more subsequent purchases within a reasonable timeframe.

In the e-commerce ecosystem, online reviews are a critical form of user-generated content that serves as both direct feedback for the firm and a purchasing guide for other consumers. Research indicates that items with positive reviews have a higher likelihood of being purchased over those with no reviews available (Biswas et al., 2021). Recognizing this influence, businesses and platforms actively encourage customers to share their experiences and thoughts on products. This user-generated content is a cornerstone of modern digital marketing, building trust and providing social proof to businesses. Trust, which then influences a customer's decision to buy, suggests that it has a longer-term effect on customer e-loyalty (Kim et al., 2009). However, the direct connection between the act of reviewing and the reviewer's subsequent loyalty is less explored. This hypothesis seeks to bridge that gap by testing whether customers who express high satisfaction through positive reviews are more likely to demonstrate loyalty through repeat purchases, thereby providing a measurable link between positive feedback and customer retention. This e-commerce platform provides a wide variety of product types, ranging from daily necessities like ‘food’, ‘drinks’, to big purchases that people only buy when needed, such as ‘computer_accessories’ or ‘console_games’. Therefore, it would be necessary to categorize product types into each group to figure out the reasonable timeframe for it to be considered frequently re-purchased.

H2: The volume of customer purchases on the Olist platform is significantly higher during the holiday season, defined as the months of November and December.

Seasonal fluctuations in sales are a well-documented phenomenon in retail. The holiday season, in particular, represents a massive surge in consumer spending, with global e-commerce holiday sales hitting a record-breaking $1.2 trillion in 2024. Specifically, the report mentioned 2 main events: Cyber Week and Black Friday to be considered as ‘holiday shopping season’ (Schwartz, 2025). In addition, a study by Salma et al. has shown a similar pattern for Brazilians who live in São Paulo, with consumption of e-commerce when this occasion arises. Specifically, the authors mentioned that holiday e-commerce contributed to around 32% of the total orders in 2017, and the $CO2$ emissions were significantly higher at around 2 times more than those of regular e-commerce. This signal for the trends being well-established not only in developed markets but also in certain areas of Brazil, it is crucial to validate its universality in the distinct economic and cultural context of all of Brazil, since the heterogeneity of services, infrastructure, access, or lack of resources contained in each city neighborhood differentially impacts residents' income conditions. (de Sousa Filho et al., 2022). And in Brazil, particularly, the situation is often especially stark (Ferreira & Ravallion, 2008). This hypothesis specifically seeks to validate whether the seasonal sales volume fluctuations are empirically observable and measurable within the shopping holiday season timeframe, ranging from November to December, compared to the rest of the year, based on the data from the Brazilian Olist marketplace.

H3: Consumer product preferences differ by geographic location, specifically, customers in urban areas purchase a higher proportion of hedonic products.

A significant factor influencing consumer preference is the distinction between hedonic and utilitarian shopping motivations, which often correlates with geographic location. Utilitarian consumption is driven by practical needs, where purchasing decisions prioritize functionality, necessity, and economic value. Research by Sinha and Verma (2019) suggests that consumers in rural areas often exhibit stronger utilitarian motivations, likely due to economic factors and a focus on practical necessities. In contrast, hedonic consumption is motivated by the pursuit of pleasure, experience, and excitement, leading to demand for goods like luxury items, high-fashion, or cutting-edge technology. Therefore, it is understandable that the urban environments, which are characterized by higher population density and wages, tend to foster elevated hedonic motivations. Results from Kim's study (2006) showed that inner-city consumers have higher hedonic motivations for shopping compared to non-inner-city consumers. Given the key characteristics of this niche market, this study will investigate whether this established urban-hedonic and rural-utilitarian framework holds true within the emerging e-commerce market of Brazil. This hypothesis will test whether this classic urban-hedonic and rural-utilitarian divide holds true within the complex and diverse Brazilian market. The approach for this hypothesis would be utilizing the categorization from essentials to non-essentials of the H1 hypothesis, and clustering the regions (through variable 'customer_zip_code_prefix', 'customer_city', and 'customer_state) into the 5 major regions: Southeast, Central West, South, North, and Northeast. The next steps in the procedure would be to inspect the relationship between the volume of sales in hedonic and utilitarian goods versus whether the regions where they live are either rural or urban. While there are no clear data on which regions are rural or not, the 2022 Census from the Brazilian Government listed out the percentages of urban populations of these regions, and it will be a good basis for considering the urbanization of each region.

Method

Data Acquisition

This study will take a deeper look at the publicly available dataset from Olist, a Brazilian e-commerce that has 100k orders from 2016 to 2018 made at multiple marketplaces in Brazil. Olist connects small businesses from all over Brazil to channels without hassle, and thus, those merchants would be able to provide a listing of items and sell their products through Olist and ship them directly to the customers using the Olist logistics partner. The data was provided by Olist itself and was anonymised since it was the real commercial data, and leaking this to the public could lead to plenty of legal problems (Olist & André Sionek, 2018). The dataset includes 9 comma-separated values (CSV) files, of which this study will be using 6 of those to tackle the hypothesis. Additionally, the 2022 Demographic Census of Brazil, which is publicly available on IBGE, is also included in this study.

Data Cleaning

In this section, the paper will go into the procedure to compile the necessary dataset after obtaining it. Note that the process had been completed mostly through Python and Excel (to see the Python code and raw data files, click here).

H1: Customer's loyalty and review scores

After importing the necessary data into Python for this hypothesis, which is the "olist_customers_dataset", "olist_orders_dataset", "olist_order_items_dataset", and "olist_order_reviews_dataset", an overview of the "customer_unique_id" (which indicates every unique customer) was needed to see what the pattern of customers who repurchased on Olist is in the "olist_customers_dataset". Moving on, those "customer_unique_id" that only appeared once in the list were excluded in order to filter out those who did not make a repeated purchase. Since the final result requires the needs of the "seller_id" to match up with the "customer_unique_id" to see if the customer has gone back to that original shop to buy more items, as well as the ratings on each of those orders, data merging is a must. The "olist_customers_dataset" that dropped those "customer_unique_id" appeared once will be used at the start of this merging section, in which it is merged by using the key column (the same column that appears on the dataset that was used as an identifier). Following that, the dataset will be merged with the "olist_orders_dataset" using the key column "customer_id" and then merged with "olist_order_items_dataset" using the "order_id" key. Finally, a merge with "olist_order_reviews_dataset" is also required through the key column "order_id" to complete linking up every variable needed for this hypothesis.

The next step after merging the dataset is to count the number of orders of each unique identifier for each customer-seller pair. There were many repeated “order_id” on the list due to the nature of this dataset, as it stated that for each item in the same order, a new row is created, but the “order_id” is still kept the same. Thus, in order to accurately measure loyalty and distinguish it from one-time bulk purchases, there are certain criteria that need to be fulfilled: “order_id” that repeats itself will be excluded as a duplicate, and only the first instance will be kept, as it is likely just a listing of the bulk order. Secondly, the orders that are from the same customer-seller pair occurring on the same date were treated as a single transaction instance since it is very unlikely that customers would receive the product on that same day and order it again; thus, the “order_purchase_date” variable was created from the “order_approved_at”, specifically by removing the time in order for an accurate comparison. Finally, the duplicates based on the mentioned criteria were removed to filter for genuine subsequent visits.

In addition to those customers who were repeat buyers, a second data frame for those who only purchased once is also needed for the full picture. The data merging steps would be similar to the data frame that has repeated purchases from customers, but this time, not filtering anything out prior to the merging process. After merging 4 of the CSV files together, the next step would be counting the unique number of “order_id” that each pair of customer-seller has, as this would only look at the number of orders that each of the pair has made. Those customer-seller who has the number of unique “order_id” more than 1 means that the customers have re-purchased at that shop again, and thus were dropped from the data frame.

Finally, the data was aggregated by the customer-seller pair to calculate the average review score (ranging from 1 - 5) and the total count of distinct orders, and merged the repeated and non-repeated dataframes together, creating a final dataset suitable for regression analysis.

H2: Number of customers' repeated purchases in the same shops

To address the second hypothesis regarding seasonal fluctuations and product types, the “olist_orders_dataset” served as the primary temporal reference, with the addition of 2 new datasets: “olist_products_dataset” and “product_category_name_translation”. First, the “order_purchase_timestamp” variable was converted into a datetime object to extract and format the data into ‘YYYY-MM’ periods, allowing for a monthly aggregation of sales volume. The order data was subsequently merged with the “olist_order_items_dataset” via the “order_id” key, and then with the “olist_products_dataset” using the “product_id” key. To classify these purchases, a binary classification was applied to the “product_category_name_translation”, labeling them as either hedonic (1) or utilitarian (0) based on research using online search engines. Irrelevant columns such as product dimensions and descriptions were dropped to streamline the data frame. The next step would be to match the primary dataset with “product_category_name_translation” in order to classify each of the products in customers’ orders as either hedonic or utilitarian. Then, the data was grouped by “order_purchase_month” to calculate the mean hedonic ratio and the total volume of orders per month, preparing the data for logistic regression analysis. Finally, the label “is_shopping_holiday” is created to categorize which months are the shopping holiday season (categorized as 1 in a binary variable), which were November and December for Christmas, May for Mother’s Day, August for Father’s Day, June for Valentine’s Day, and October for Children’s Day (Tech in Brazil, 2015). The rest of the months in the year would be coded as 0, and as for the hedonic ratio variable, it was coded as follows: the hedonic ratio was calculated as a continuous variable ranging from 0 (indicating purely utilitarian purchases) to 1 (indicating purely hedonic purchases).

H3: Geographic influence on product preference

The third hypothesis required a spatial analysis of consumer behavior, specifically examining the relationship between urbanization and product preference. The procedure began by identifying the unique “customer_state” from the customer dataset, noting that the data included 27 Brazilian states, which is the same number of states mentioned in the Brazilian Census Data. Similar to the previous procedure, the customer data was merged with order, item, and product datasets to link specific transactions to geographic locations. Unnecessary columns related to shipping and approval timestamps were removed. This merged dataset was then combined with the hedonic category binary variables established in the H2 procedure. The data was grouped by “customer_state” to calculate the aggregated hedonic consumption ratio and total order count for each state. To complete the dataset for analysis, these state-level consumption metrics were merged with external urbanization data derived from the IBGE 2022 Census, linking the percentage of urban population in each state to their respective consumption patterns. The hedonic ratio variable was similar to H2. Finally, a variable named “order” was created to label each of the states with a number for further analysis.

Defining Outliers

Before the initial analysis begins, a process of inspecting the dataset to see if there are any outlier variables that need to be considered has been run. This process included the Mahalanobis distance (leverage), studentized deleted residuals (discrepancy), and Cook’s Distance (“Global influence”).

Results

The dataset was first screened for outliers using Mahalanobis distance, Studentized deleted residuals, and Cook’s distance. No significant outliers were identified for H1 or H3 that warranted removal.

H1: Customer Reviews and E-Loyalty

Table 1 reveals a statistically significant correlation ($r = .01$, $p < .01$) between review scores and the number of orders. However, it is crucial to interpret this with caution. In extremely large datasets ($N = 97967$), even a virtually weak positive relationship can appear "statistically significant" simply because of the sample size (see Table 1-2). Practically speaking, a correlation of .01 is negligible. Therefore, this significant correlation should not be taken lightly. With the filtered population (the subset of buyers who made more than one purchase), there was no significant correlation between the review scores and the number of orders ($r = .05$, $p = .20$)

A Poisson regression was conducted to predict the number of subsequent orders based on the average review score provided by the customer. The initial analysis included the full dataset of all customers ($N = 97967$), in which the relationship between average review score and the number of orders was not statistically significant (B = 0.001, $p = .81$), meaning that the review scores did not predict the increasing number of orders. Thus, a frequency analysis was made, which revealed a substantial skew in the data, with 99.4% of customers ($n = 97412$) making only a single purchase. To address this abundance of one-time buyers and investigate if reviews predict loyalty among those who do return, a second Poisson regression was performed on the subset of customers who made more than one purchase ($n = 555$). In this filtered model, the relationship between average review score and the number of subsequent orders remained non-significant (B = 0.008, $p = .77$).

Given the lack of significance in the regression models, an exploratory post-hoc one-way ANOVA was conducted on the repeat customer subset ($n = 555$) to test if average review scores differed significantly across specific order-count groups (2, 3, 4, or 7 orders). The ANOVA indicated no significant differences in review scores between customers with varying levels of repeat purchases, F (3, 551) = 0.88, p = .45. Furthermore, the result showed that the means of the review scores regarding the number of orders stay relatively similar to each other, with M = 4.08 (SD = 1.34) for the full population compared to M = 4.23 (SD = 1.04) of the filtered population. These results collectively indicate that review scores are not a significant predictor of customer loyalty or repeat purchase frequency.

Table 1

Descriptive Statistics for Full Population (Hypothesis 1)

Variable	*M* (*SD*)	1.
1. Average review score	4.08 (1.34)	--
2. Number of orders	1.01 (0.08)	.01

Note. N = 97967. Correlation significant at the p < .01 level.

Table 2

Descriptive Statistics for Filtered Population (Hypothesis 1)

Variable	*M* (*SD*)	1.
1. Average review score	4.23 (1.04)	--
2. Number of orders	2.05 (0.32)	.05

Note. N = 555.

H2: Seasonality and Holiday Shopping

When screening for outliers using Cook's Distance and studentized deleted residuals, two months were identified as outliers based on these metrics. Upon further inspection, these months shared a common characteristic of having a very low total order volume (< 6 total orders). A third month also met this low-volume criterion and was subsequently removed to ensure data quality. The final analysis was conducted on the remaining 21 months (N = 21).

A linear regression was performed to determine if the hedonic purchase ratio differed significantly during the holiday shopping season. The model was not statistically significant, F (1, 19) = 0.54, p = .47, accounting for only 2.8% of the variance. Although the mean hedonic ratio was slightly higher during holiday months (M = 0.58, SD = 0.05) compared to non-holiday months (M = 0.57, SD = 0.04), this difference was not statistically significant.

Table 3

Descriptive Statistics for Hypothesis 2

Variable	*M* (*SD*)	1.
1. Hedonic ratio	0.57 (0.04)	--
2. Is shopping holiday?	0.43 (0.50)	.16

Note. N = 21.

H3: Geographic Location and Product Preference

There is a strong, negative correlation (r = -.47) between urbanization and the hedonic ratio (Table 4). This means that as a state becomes more urbanized (higher %), the proportion of hedonic (luxury) purchases goes down.

A linear regression analysis was performed to test the hypothesis that customers in more urbanized areas purchase a higher proportion of hedonic products (N = 27). The model was statistically significant, *F* (1, 25) = 7.27, *p* = .01, explaining 22.5% of the variance in the hedonic ratio. The regression results, along with the weak negative relationship, contradicted the hypothesis: urbanization percentage was a significant *negative* predictor of the hedonic ratio (B = -0.002, *p* = .01). Therefore, the dataset suggests that the higher urbanization level predicts the lower proportion of hedonic products purchased.

Table 4

Descriptive Statistics for Hypothesis 3

Variable	*M* (*SD*)	1.
1. Hedonic ratio	0.59 (0.34)	--
2. Urbanization rate (%)	83.64 (7.57)	-.47

Note. *N* = 27. Correlation significant at the *p* < .005 level.

Discussion

This study utilized the Olist Public Dataset to investigate three critical dimensions of the Brazilian e-commerce market: the relationship between customer satisfaction and loyalty, the impact of seasonality on purchase behavior, and the influence of geographic urbanization on product preferences. The findings offer a nuanced view of the Brazilian digital marketplace, challenging several traditional marketing assumptions. When it comes to customer loyalty and satisfaction, while previous articles such as Kim et al. (2009) mentioned that positive customer reviews would predict higher e-loyalty, this study's analysis revealed no significant relationship between review scores and repeat purchase behavior in either the general population or the subset of repeat buyers. Furthermore, the descriptive data highlighted an overwhelmingly transactional market, with 99.4% of customers making only a single purchase. This lack of loyalty suggests that the Olist platform may function more as a marketplace for price-sensitive or specific-need discovery rather than a brand-building ecosystem. In addition, it could also be caused by the fact that this dataset was collected in 2018, an era in which online e-commerce usage was less dominant, and the value of online shopping was not on par with physical stores. Therefore, customers may attribute their satisfaction to the e-commerce platform rather than the seller, and while this does not necessarily mean reducing the incentive to return to the same specific store, it helps the consumer understand how useful the platform can be. The results imply that for small merchants on aggregated platforms, high ratings alone are insufficient to guarantee customer retention without additional engagement strategies.

Seasonality and holiday consumption were also examined by looking at whether the "holiday shopping season" drove significant shifts in the consumption of hedonic goods. While global trends suggest a massive surge in holiday spending, the regression analysis found no statistically significant difference in the hedonic purchase ratio between holiday and non-holiday months, although the mean hedonic ratio was a bit higher during holidays compared to non-holidays. This could happen due to several reasons, one of which is that Brazilian does not fully perceive e-commerce as one of the main methods to go shopping during the holidays, possibly due to their habits or the pricing was not as well offered as in physical stores. Additionally, it is possible that while total volume has a slight increase during holidays, the proportion of hedonic to utilitarian goods remains balanced because consumers utilize holiday shopping events like Black Friday to stock up on both necessities and luxuries equally.

The most striking findings from the third hypothesis which predicted that urban areas would favor hedonic products while rural areas would favor utilitarian ones. The results significantly contradicted this expectation, showing a negative relationship: as urbanization increases, the proportion of hedonic purchases decreases. This could be explained by several factors: in highly urbanized Brazilian states (e.g., São Paulo, Rio de Janeiro), consumers have immediate physical access to luxury malls and high-end retailers, allowing them to purchase hedonic goods offline. Conversely, consumers in rural or less urbanized regions often lack access to physical specialty stores. Therefore, they may rely more heavily on e-commerce to acquire hedonic items that are unavailable in their local physical markets. In addition, for necessity items, shops would likely have a warehouse that is placed near central cities, which could result in faster delivery time compared to the more rural areas and thus, drives the incentive for the urbanized population and vice versa. Furthermore, it might be due to the lack of knowledge about the item's characteristics (price, model, quality, etc.) among the population that could lead to rural people buying more hedonic goods online than urban people.

However, this study is not without limitations. First and foremost, the dataset spans 2016–2018, a period of economic volatility in Brazil, which may have influenced spending power. Additionally, when coding the first hypothesis, the study only refers to ordering on the same day as a singular order since the item would not be likely to arrive on the same day, but this could extend to 2 to 3 days, and thus, counting those orders on the next 2-3 days as subsequent orders would not be sufficient. Similarly, another coding limitations appear in the second hypothesis, specifically in the holiday shopping coding session, in which the source found to classify the "is_shopping_holiday" variable is not a peer-reviewed article, so the credibility is questionable. Furthermore, using review scores as the sole proxy for satisfaction may miss those who were satisfied but did not complete a rating or review. Finally, an e-commerce platform might not be the ideal place to test out these hypotheses, as it might not as convenient at the time to buy online as buying it in person due to many reasons: the popularity of e-commerce, the technology was not well developed at the time, lack of access to the platform because of poverty, etc.

Future research could investigate the distinction between platform loyalty (returning to Olist) versus physical shopping frequency to better understand the drivers of retention in the gig-economy marketplace. Furthermore, with the rise of technology and the epidemic in the timeline, the dataset in the following years would grow exponentially, thus providing better evidence for future studies to base on. Overall, these findings provide actionable insights for managers operating in emerging markets. Specifically, the low repurchase rate suggests that acquisition costs must be kept low, as the "lifetime value" of a customer on this specific platform is limited at the time. Moreover, this information can be provided to the sellers in order for them to better understand how e-commerce markets work, and try to come up with solutions for customers to return to their store with all kinds of incentives: coupons, sales, buy one get one, etc. Not only would it keep customers from coming back, but it also serves as a reason for sellers to come and sell at Olist. Finally, the geographic data suggests a targeted strategy for logistics and marketing. When it comes to hedonic/luxury goods, the organization should potentially target rural and semi-urban areas where local supply is low, rather than oversaturating urban centers.

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